

Petrol, diesel fuel and jet fuel demand in South Africa: 1998-2009

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Abstract

We employ the autoregressive distributed lag (ARDL) model and the automated general-to-specific approach to re-estimate price and income elasticity for petrol and, for the first time, diesel and jet fuel, in South Africa. By using a quarterly dataset, we build adequate and parsimonious econometric models for the 1998-2009 period, avoiding the problem of structural breaks in longer sample periods – an issue which previous studies built on annual data ignore. Our price elasticity estimates for petrol demand is consistent with previous South African research, but we find income elasticity to be significantly higher than previously estimated. Diesel and jet fuel demand exhibits low price elasticity and very high income elasticity. Finally, the speed with which demand respond to long-run disequilibria is found to be much slower for petrol than for either diesel fuel or jet fuel.

Keywords: petrol demand, diesel, jet fuel, petrol, South Africa, bounds test

JEL classification: C22, C52, C53, R41

1. Introduction

We present long-run estimates of the price and income elasticity of petrol, diesel fuel, and jet fuel demand in South Africa for the period 1982Q1-2009Q3 as well as for the shorter recent period 1998Q1-2009Q3. Previous fuel demand studies for South Africa, including recent work of Akinboade, Ziramba and Komo (2008), are based on fuel consumption and price data of an annual frequency in order to obtain long enough sample periods that avoid econometric problems associated with short samples. However, as opponents of time-series demand models have noted, these models may not account adequately for changes in consumer preferences, technologies and institutional change. An alternative approach may be to study cross-sectional data, but cross-sectional models are not feasible owing to a lack of appropriate survey data in South Africa. We therefore also follow a time series approach, but based on a dataset of quarterly frequency, which allow enough data points to study a recent sample period starting in 1998. The shorter sample period enables us to assess whether fuel demand drivers have changed under a new institutional environment (following the dismantling of Apartheid in the early 1990s) and with the availability of fuel-efficient technologies. Recently, Theron (2008) showed the implications of incorrect fuel demand estimates in South African competition policy investigations, while elasticity estimates have also features in the national debate on proposed petroleum pipeline tariff adjustments (National Energy Regulator of South Africa 2010).

Apart from these practical motivations, we also make a methodological contribution by explicitly adopting a general-to-specific modelling approach in fitting autoregressive distributed lag models for fuel demand for South Africa. While Akinboade et al (2008) employ a manual reduction approach in their annual South African models, we follow the automated search algorithm developed by Hendry and Krolzig (2001) and embodied into the *OxMetrics* software package (Doornik 2008). The automated algorithm employs a diversity of reduction paths, reducing potential error arising from pursuing any one particular path.

2. Literature review

The literature on energy demand models distinguishes between theory-driven and empirically-driven approaches to demand modelling. Empirical approaches include a range of statistical models and econometric models. Statistical models include autoregressive specifications and crude smoothing procedures and appear to outperform more sophisticated econometric and theoretical models in forecasting (Li, Rose et al. 2010). Nevertheless, for policy analysis and retrospective analysis, the

modelling of econometric relationships remains useful. Econometric models of fuel demand boast a range of methods, including application of the recent bounds-testing approach developed by Pesaran, Shin and Smith (2001) – De Vita et al (2006) and Akinboade et al (2008) are applications in the South African context. However, even these newer econometric models face serious challenges in dealing with structural change, a feature likely to be increasingly important in decades to come. Theoretical approaches, such as the partial adjustment model, have been applied recently to allow for changes in habits and structural breaks (Breunig and Gisz 2009). The models seem promising, but appear to be less useful for forecasting purposes.

Fuel demand in South Africa has been extensively investigated over the past two decades. Theron (2008) provides a useful summary of earlier work as well as recent private-sector estimates, to which we add some additional results from the academic literature, as shown in Table 1. The price elasticity estimates for petrol demand are generally around -0.5 and for diesel demand around -0.1. Income elasticity of petrol demand are estimated at 0.4 (with the exception of one estimate of 1.0) while the income elasticity of diesel appears to be above 1.0.

Table 1: Estimates of price and income elasticity of South African fuel demand

Authors	Sample period	Price elasticity	Income elasticity
Cloete and Smit (1988)	1970-1983	-0.24 (petrol, short-term) -0.37 (petrol, long-term)	0.43 (petrol)
Bureau for Economic Research (2003)	n.a.-2003	-0.21 (petrol, short-term) -0.51 (petrol, long-term) -0.18 (diesel, short-term) -0.06 (diesel, long-term)	n.a.
Akinboade et al (2008)	1978-2005	-0.47 (petrol, long-term)	0.36 (petrol, long-term)
Theron (2008)	1984-2004	-0.19 (petrol, short-term)	0.1 (petrol, short-term)
BER model		-0.62 (petrol, long-term) -0.1 (diesel, long-term)	1.0 (petrol, long-term) 1.36 (diesel, long-term)
Theron (2008)	n.a.	-0.24 (petrol)	0.38 (petrol)
Econometrix model		-0.14 (diesel)	1.47 (diesel)

The striking message from Table 1 is that the more recent estimates by Akinboade et al (2008) show very little difference from the first formal results in Cloete and Smit (1988): this is potentially inconsistent with evidence of improvement in fuel technology and changes in fuel consumption

behaviour over the past two decades, which could have altered the price and income elasticity of fuel demand. We therefore re-investigate South African fuel demand to study the potential impact of structural breaks using a more recent shorter sample period of a quarterly frequency.

3. Methodology

The paper employs the autoregressive distributed lag (ARDL) model proposed by Pesaran et al (2001) to model fuel demand, coupled with an automated general-to-specific (GETS) reduction strategy. Modelling therefore starts with estimating an unrestricted ARDL model of lag order p :

$$\Delta Q_t = \beta_0 + \sum_{i=1}^p \beta_{1,i} \Delta Q_{t-i} + \sum_{i=1}^p \beta_{2,i} \Delta P_{t-i} + \sum_{i=1}^p \beta_{3,i} \Delta Y_{t-i} + \alpha_1 Q_{t-1} + \alpha_2 P_{t-1} + \alpha_3 Y_{t-1} + \gamma' Z_t + \varepsilon_t$$

where, in period t , Q_t is log fuel sales, P_t is log fuel price, Y_t is log income and Z_t is a vector of dummy variables dealing with data outliers.

Following a GETS procedure, we inspect whether the model is congruent with both data and theory: we investigate whether the signs of different parameter estimates are consistent with the predictions from theory (for example, overall negative sign for price elasticity and positive sign for income elasticity) and then run a batch of misspecification and diagnostic tests on the residuals $\hat{\varepsilon}_t$ (including tests for normality, heteroscedasticity, remaining autocorrelation and the Ramsey RESET test for specification error). If these tests are passed, the model is known as a general unrestricted model (GUM).

The GUM is not a parsimonious model and may contain, for example, irrelevant lagged variables that may contaminate the long-run parameter estimates and lead to a less robust model. Consequently, we employ an automated GETS search algorithm to reduce the GUM to a specific model. The algorithm chooses a number of starting points and, for each path, employs a step-wise reduction strategy to omit statistically insignificant variables provided information loss is limited (information loss is measured by change in the maximized log-likelihood value). The results of the multiple paths are then unified in a single model, on which the same step-wise reduction procedure is repeated until the model arrives at a single parsimonious model – known as the specific model.

The specific model allows us, firstly, to test for the existence of a long-run relationship and, secondly, to derive long-run elasticity estimates. Pesaran et al (2001) show that a test for the existence of a long-run relationship involves testing the hypothesis that $\alpha_1 = \alpha_2 = \alpha_3 = 0$ against two-sided alternatives. These authors then suggests a bounds test approach, according to which the critical F-statistic is compared with two critical bounds, an upper value associated with the condition where all of P , Q and Y are $I(1)$, i.e. contain unit roots, and a lower value where all of P , Q and Y are $I(0)$, i.e. are stationary. Values falling below the lower boundary indicate the absence of a systematic relationship, while values exceeding the upper boundary confirm such a relationship. Where the test statistics fall between the two critical values, it is necessary to test for unit roots in the individual series. If the series are all integrated, the upper bound is the critical value. Where all series are found stationary, the lower bound is the critical value. For a combination of stationary and non-stationary variables the test is inconclusive if the test statistic falls between the critical bounds. The latter is not common and the bounds test approach therefore avoids (or, at least, significantly reduces) the need for pre-testing the series for unit roots.

Once the existence of a long-run relationship is established, it is straightforward to calculate estimates for the long-run price and income elasticity of fuel demand:

$$\hat{\theta}_{price} = \frac{\hat{\alpha}_2}{\hat{\alpha}_1}$$

$$\hat{\theta}_{income} = \frac{\hat{\alpha}_3}{\hat{\alpha}_1}$$

The parameter estimate $\hat{\alpha}_1$ is the so-called speed of adjustment parameter, if all series are non-stationary: it shows the speed at which the ΔQ_t will respond to any long-run disequilibria. For example, if the speed of adjustment parameter is small the long run plays a less important role in the quarter-to-quarter behaviour of fuel consumption and short-run factors may be more important.

4. Data description

4.1 Variables and data sources

The first step in economic modelling is the identification of the parameters of interest and the collection of data on variables that will enable estimates of these parameters. In general, the demand for any good depends on various factors, including its own price, income, and prices of substitutes and complements. The literature has also emphasized the importance of accounting for a plethora of additional demand-

shift factors, including preferences, technology and institutional change. While all these variables would provide a rich model of fuel demand, the empirical estimation of such a function is challenging. Specifically, accounting for changes in the underlying tastes and preferences of consumers as well as for changes in the institutional environment is a difficult task. This paper improves on current research by considering a shorter quarterly dataset that may be less exposed to structural breaks, while continuing to focus only on price and income forces due to data constraints. Table 2 reports the data sources used in the econometric analysis:

Table 2: Variables and data sources

Variable	Source	Description
Petrol sales	South African Petroleum Industry Association (SAPIA) (up to 2008), Department of Minerals and Energy (DME)	Petrol sales in millions of litre, 1982Q1-2009Q3
Diesel fuel sales	SAPIA (up to 2008), DME	Diesel sales in millions of litre, 1982Q1-2009Q3
Jet fuel sales	SAPIA (up to 2008), DME	Jet fuel sales in millions of litre, 1994Q1-2009Q3
Petrol price	SAPIA	Retail coastal pump price of 95 octane petrol in Rand, 1982Q1-2010Q1
Diesel price	SAPIA	Retail coastal pump price of 0.05% sulphur diesel in Rand, 1982Q1-2010Q1
Oil price	South African Reserve Bank (SARB)	Quarterly Brent crude oil (spot) in US dollars (data series KBP5344M), 1980Q3-2009Q4
General price level	SARB	Private consumption deflator, base year 2005, 1982Q1-2009Q4
Income	SARB	Household disposable income in millions of Rand, base year 2005 (data series KBP6246L), 1982Q1-2009Q4 Real gross domestic product in millions of Rand, base year 2005 (data series KBP6006D), 1982Q1-2009Q4
Rand dollar exchange rate	SARB	Rand dollar exchange rate (data series KBP5339M), 1982Q1-2009Q4

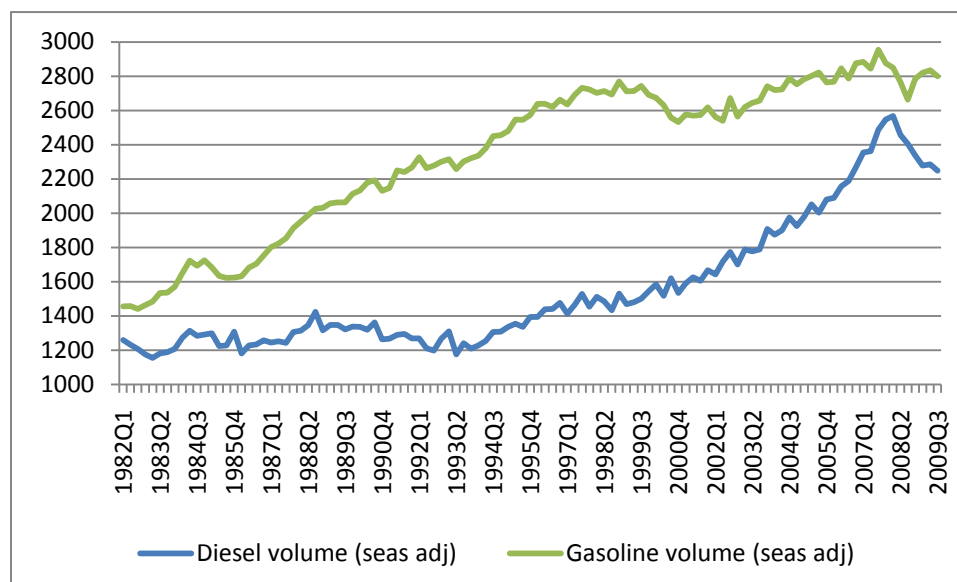
As far as prices are concerned, we use the real retail price of petrol and diesel fuel and the real oil price in South African currency (rand) for jet fuel (we do not have actual jet fuel prices available). For income, we use real disposable income for petrol and real gross domestic product (GDP) for diesel fuel and jet fuel. The difference is motivated from previous South African research, which find a better fit for

disposable income than GDP in petrol demand functions (Theron 2008). Before proceeding to the formal modelling, we present a brief descriptive analysis of the fuel consumption data in order to highlight structural breaks.

4.2 Structural breaks

This study is motivated by the need to reassess price and income elasticity on the basis of a more recent and quarterly dataset that is less prone to structural breaks than longer and annual data employed by previous researchers. Given the high levels of seasonality in quarterly fuel consumption data, we first seasonally adjust the series using the standard X-12 procedure of the US Census Bureau. Figure 1 reports seasonally adjusted petrol and diesel fuel volumes.

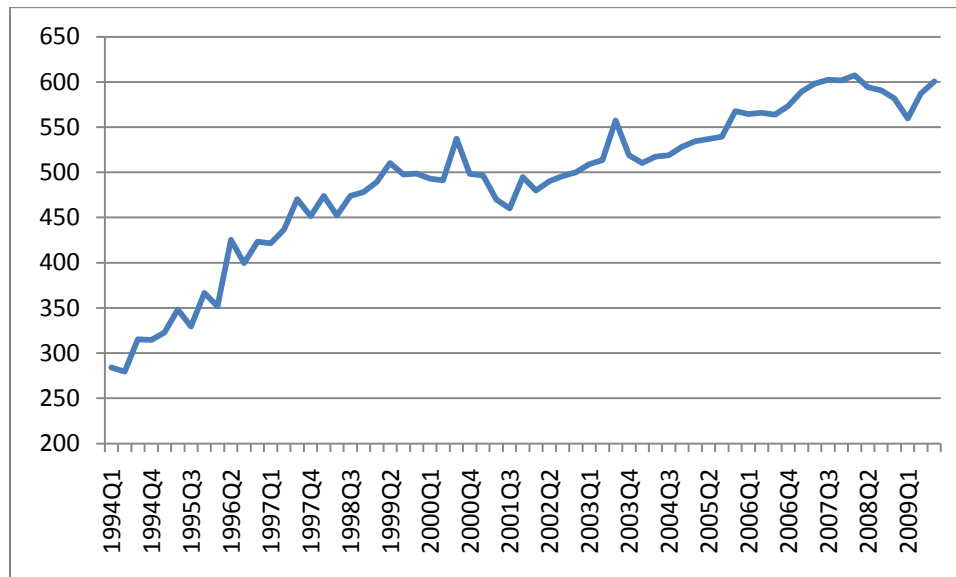
Figure 1: Seasonally adjusted petrol and diesel fuel sales in South Africa (millions of litre), 1982Q1-2009Q3



The figure suggests a structural break around 1998 in petrol volumes: before 1998 a strong time trend is visible, but none after 1998. At around the same time, or perhaps slightly earlier, diesel fuel volumes appear to accelerate strongly relative to the previous sideways movement. In Figure 2, jet fuel volumes behave similarly (although we only have data available from 1994 onwards): volumes experience strong growth up to around 1998 but subdued growth subsequently. We therefore model demand for petrol and diesel fuel both for the entire sample period from 1982Q1-2009Q3 and the shorter period of 1998Q1-2009Q3. Jet fuel is modelled only for 1998Q1-2009Q3 due to data constraints. The longer

sample period provides a standard against which to assess earlier estimates, including the Akinboade et al (2008) results, while the shorter recent sample period provides an indication of how elasticity estimates may have changed in recent years.

Figure 2: Seasonally adjusted jet fuel sales in South Africa (millions of litre), 1994Q1-2009Q3



4.3 Lag order and dummy variables

Lag structure is an important feature of a time series demand model, allowing for a protracted impact of price or income changes on fuel consumption. However, lag orders that are too large will produce bias in econometric results. Akinboade et al (2008) use a lag order of three years in their annual model of South African petrol demand. Translated to quarterly frequency, this implies a lag length of twelve quarters, which seems very long. We start each of our unrestricted fuel demand models with a lag length of four quarters. The specific models that emerge from the GETS reduction process usually contain lags of a first or second order, which suggests that four lags are not restrictive.

In addition to the lagged variables we also include occasional dummy variables to account for data outliers, listed in Table 3. The dummy variables significantly improve the stability of the GUMs.

Table 3: Dummy variables in fuel demand models

Petrol model	Diesel fuel model	Jet fuel model
2002Q2	1992Q4-1993Q1	2003Q3
	1993Q2	2000Q3
	1988Q4	

5. Results

This section presents the regression results for the different models. As argued, our approach involves, firstly, generating a GUM and, secondly, a specific model. Here we only present the specific model results. In each case, the GUM passed all data misspecification tests and was found congruent with theory. The results are available upon request. The regression results is then followed by misspecification tests, graphs related to parameter stability, then the bounds test results and finally a table reporting the long-run estimates of price and income elasticity.

5.1 Petrol

Table 4 present the petrol demand models for both the longer and shorter sample periods:

Table 4: Specific models for petrol demand (dependent variable ΔG_t)

1982Q1-2009Q3		1998Q1-2009Q3	
Regressor	Coefficient (Standard error)	Regressor	Coefficient (Standard error)
ΔG_{t-1}	-0.24 (0.07)	ΔG_{t-1}	-0.30 (0.11)
		ΔG_{t-4}	-0.28 (0.12)
$\Delta P_{G,t}$	-0.17 (0.02)	$\Delta P_{G,t}$	-0.14 (0.03)
$\Delta P_{G,t-4}$	0.06 (0.02)		
$\Delta Y_{DispInc,t}$	0.21 (0.05)	$\Delta Y_{DispInc,t-2}$	0.62 (0.24)
G_{t-1}	-0.20 (0.03)	G_{t-1}	-0.25 (0.10)
$P_{G,t-1}$	-0.11 (0.02)	$P_{G,t-1}$	-0.13 (0.03)
$Y_{GDP,t-1}$	0.16 (0.03)	$Y_{DispInc,t-1}$	0.21 (0.05)

R^2	0.54	R^2	0.67
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A range of misspecification tests, reported in Table 5, confirm the adequacy of both models at a 5% significance level. However, as expected, the model based on the longer sample period fails the RESET, which likely picks up the impact of a structural break on the functional form of the regression.

Table 5: Misspecification tests for petrol demand models

	1982Q1-2009Q3	1998Q1-2009Q3
Test name	Test statistic (probability)	Test statistic (probability)
AR (1-4) test	2.16 (0.06)	1.56 (0.21)
ARCH (1-4) test	0.85 (0.50)	0.52 (0.72)
Normality test	4.26 (0.12)	5.67 (0.06)
Heteroscedasticity test	0.28 (0.99)	0.72 (0.74)
Ramsey RESET	4.16 (0.02)	0.52 (0.60)

It is possible to further investigate the stability of the two petrol demand models using recursive estimation: this method estimates the same model on smaller sub-sample periods to assess the stability of the parameter estimates. Figure 3 reports recursive parameter estimates for price and income elasticity for the model based on the longer sample period.

Figure 3 suggests significant instability in both parameters during the mid-1980s and stability afterwards. Figure 4 replicates the exercise for the petrol model based on the more recent shorter sample period and confirms that this model produces more robust parameter estimates. Therefore, even if results are similar for the two models, we are less certain about the results for the longer sample period. Put differently, we should be careful of using a single demand model spanning a long sample period, as is the practice in previous South African fuel demand research.

Figure 3: Recursive estimates for long-run price and income elasticity parameters in petrol demand models (1982Q1-2009Q3)

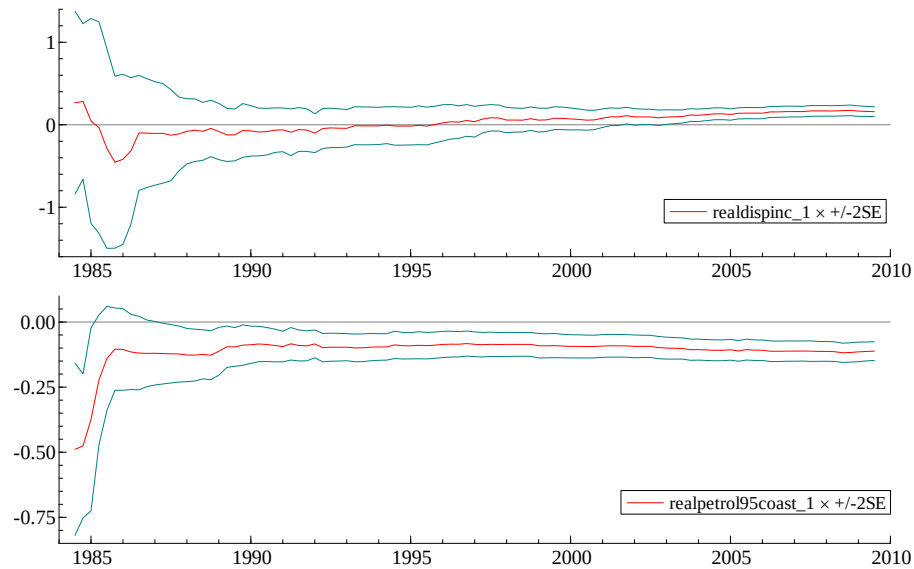
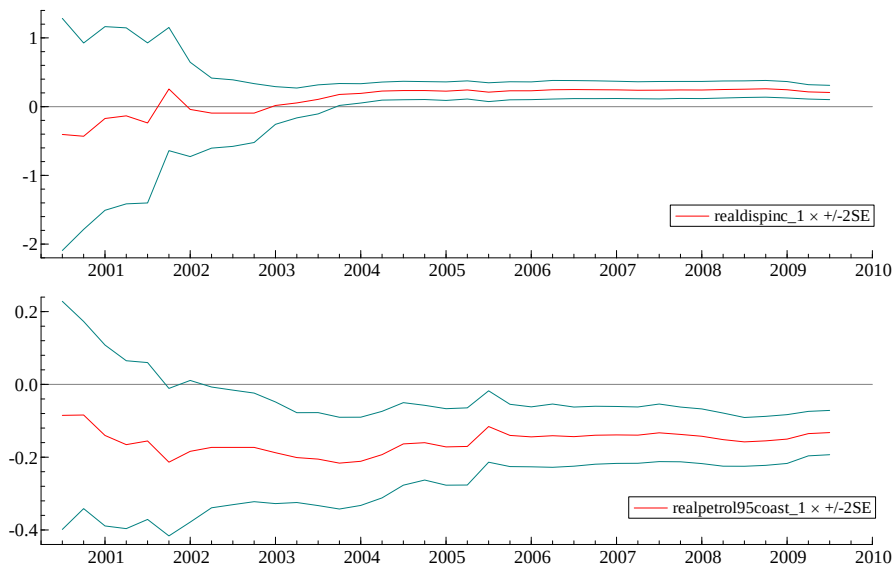


Figure 4: Recursive estimates for long-run income elasticity parameter in petrol demand models (1998Q1-2009Q3)



If one is willing to accept both specific models, one may proceed to the bounds test. Table 6 shows the results, confirming a significant long-run relationship for both sample periods.

Table 6: Bounds test results for petrol demand models

Sample period	F-statistic	5% critical bounds	
		Lower bound ($I(0)$)	Upper bound ($I(1)$)
1982Q1-2009Q3	16.27*	3.79	4.85
1998Q1-2009Q3	6.31*	3.79	4.85

* Reject null hypothesis at 5% significance level

Given confirmation of a long-run relationship, Table 7 reports estimates for the parameters in the long-run relationship. The suggested long-run price elasticity estimate from both models are around -0.55, while the long-run income elasticity is estimated at around 0.8 (see Table 7). Following Akinboade et al (2008) we estimate standard errors using Bardsen (1989).

Table 7: Long-run elasticities of petrol demand

Sample period	Price elasticity (standard error)	Income elasticity (standard error)
1982Q1-2009Q3	-0.55 (0.03)	0.78 (0.03)
1998Q1-2009Q3	-0.53 (0.18)	0.83 (0.23)

Although our price elasticity estimates for petrol demand correspond with those obtained by Akinboade et al (2008), it is not clear whether the result indicates unchanged consumer behaviour or whether the correspondence is due to chance because of parameter instability. We also find significantly higher income elasticity in both our sample periods, which suggests that consumption of petrol is actually much more sensitive to the consumer's financial position. While some previous studies also find high income elasticity, none of these studies look closely at the problem of structural change and the problem of relying on a long sample period.

Finally, the speed-of-adjustment parameter for both demand models is around 0.2 (see Table 4), which implies that it takes about five quarters for a long-run disequilibrium to be corrected. This speed is quite different from the speed suggested by previous annual data models of about five years (Akinboade, Ziramba et al. 2008). Such a protracted response is not intuitive, as is evident from simply comparing quarterly fuel consumption and price. We therefore argue that our petrol demand estimates offer an improvement over current research.

5.2 Diesel fuel

We follow a similar approach to estimate the demand function for diesel fuel in South Africa, developing a model based on both the longer and shorter sample period. We find it quite difficult to build an adequate model based on the longer sample period. Even after including dummy variables to account for extreme outliers, we do not identify a GUM congruent with both data and theory for the 1982Q1-2009Q3 sample period. For example, the model continues to exhibit positive signs for price elasticity. In contrast, we find the 1998Q1-2009Q3 GUM congruent with data and theory, showing signs consistent with theory and passing all misspecification tests. Despite the unsatisfactory results for the longer-sample GUM we also report the specific model derived from this GUM (to allow comparison with previous research on diesel fuel demand). As shown in Table 8, the longer-period GUM's problems carry over into its specific model, which includes only the long-run income elasticity without any long-run parameter for price. We therefore focus mostly on the 1998Q1-2009Q3 specific model:

Table 8: Specific models for diesel fuel demand (dependent variable ΔD_t)

1982Q1-2009Q3		1998Q1-2009Q3	
Regressor	Coefficient (Standard error)	Regressor	Coefficient (Standard error)
ΔD_{t-1}	-0.53 (0.08)	ΔD_{t-2}	0.40 (0.11)
ΔD_{t-3}	0.14 (0.05)	ΔD_{t-3}	0.56 (0.11)
$\Delta P_{D,t}$	-0.12 (0.03)	$\Delta P_{D,t}$	-0.10 (0.04)
$\Delta Y_{GDP,t}$	2.00 (0.30)	$\Delta Y_{GDP,t}$	1.63 (0.61)
$\Delta Y_{GDP,t-1}$	0.65 (0.33)		
$\Delta Y_{GDP,t-4}$	0.65 (0.28)		
D_{t-1}	-0.14 (0.06)	D_{t-1}	-0.67 (0.13)
		$P_{D,t-1}$	-0.09 (0.03)
$Y_{GDP,t-1}$	0.14 (0.06)	$Y_{GDP,t-1}$	1.01 (0.18)
R^2	0.54	R^2	0.67

Table 9 confirms the adequacy of the specific model for 1998Q1-2009Q3 and highlights some of the problems of the longer period specific model. The problems are likely the result of the significant change

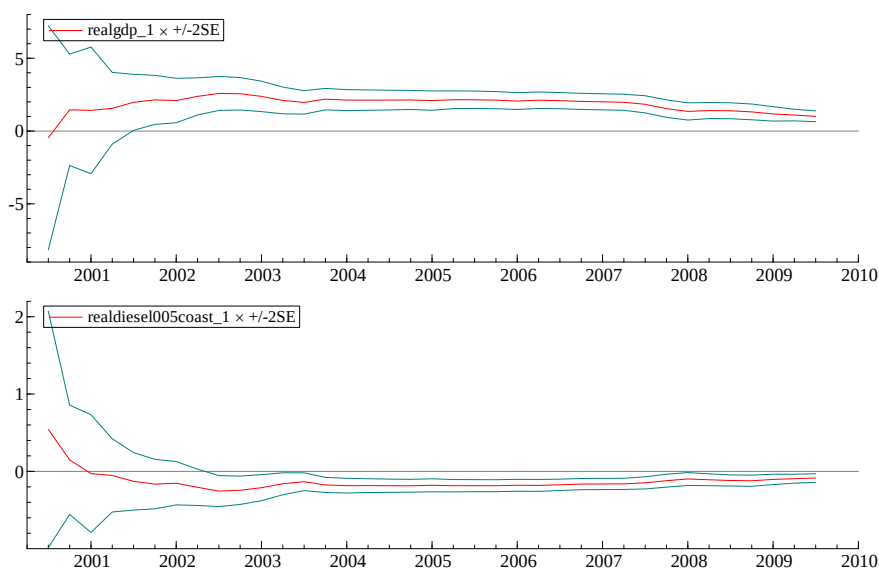
in behaviour of diesel fuel consumption during the mid 1990s – a period included in the longer sample period (refer back to Figure 1).

Table 9: Misspecification tests for diesel fuel demand specific models

Test name	1982Q1-2009Q3	1998Q1-2009Q3
	Test statistic (probability)	Test statistic (probability)
AR (1-4) test	2.25 (0.06)	0.64 (0.64)
ARCH (1-4) test	0.30 (0.88)	0.43 (0.78)
Normality test	0.10 (0.95)	1.25 (0.53)
Heteroscedasticity test	0.85 (0.63)	0.90 (0.56)
Ramsey RESET	3.56 (0.03)	0.90 (0.42)

Figure 5 reports recursive estimation results for the specific model based on the shorter sample period. The figure suggests that the estimates are fairly stable. Although confidence intervals in 2000 and 2001 are wide, these are the result of the very short subsample periods over which the first recursive estimates are derived and the graph shows subsequent estimates to be very stable.

Figure 5: Recursive estimates for long-run income elasticity parameter in diesel fuel demand model (1998Q1-2009Q3)



Both models pass the bounds test, as shown in Table 10, suggesting a significant long-run relationship between diesel fuel price, income and diesel fuel consumption:

Table 10: Bounds test results for diesel fuel demand models

Sample period	F-statistic	5% critical bounds	
		Lower bound ($I(0)$)	Upper bound ($I(1)$)
1982Q1-2009Q3	7.43*	4.94	5.73
1998Q1-2009Q3	11.51*	3.79	4.85

* Reject null hypothesis at 5% significance level

Table 11 reports the long-run elasticity estimates suggested by both models (bearing in mind that the specific model for the longer period does not have a price elasticity parameter, as noted earlier). Demand elasticities for the shorter period specific model is estimated at around -0.13 for price and 1.5 for income. These results are consistent with previous diesel fuel demand estimates for South Africa of about -0.1 for price and 1.4 for income (refer to Table 1).

Table 11: Long-run elasticities of diesel fuel demand

Sample period	Price elasticity (standard error)	Income elasticity (standard error)
1982Q1-2009Q3	-	1.01 (0.01)
1998Q1-2009Q3	-0.13 (0.04)	1.51 (0.08)

Although research on diesel fuel demand in South Africa is not ubiquitous our diesel fuel results appear to be consistent with previous findings. Specifically, the results highlight the significant role of economic growth in driving demand for diesel fuel in South Africa, much stronger than for petrol. Furthermore, even if one takes the diesel model over the longer period to be merely indicative, a case can be made that economic growth remains the most important driver of diesel fuel consumption.

5.3 Jet fuel

Jet fuel consumption data is more limited than petrol or diesel fuel data and are only available from 1994 onwards. We estimate the demand model for 1998Q1-2009Q3 to retain comparability with the other models, but also as initial results including the earlier years suggested structural breaks. We find the GUM to be congruent with data and theory and subsequently derive the specific model using the

GETS automated procedure. The regression results for the specific model are reported in Table 12 and show a very simple specification, which includes only the lagged dependent variable in addition to the long-run variables.

Table 12: Specific models for jet fuel demand (dependent variable ΔJ_t)

Regressor	Coefficient (Standard error)
ΔJ_{t-1}	-0.09 (0.03)
J_{t-1}	-0.62 (0.09)
$P_{J,t-1}$	-0.06 (0.01)
$Y_{GDP,t-1}$	0.58 (0.08)
R^2	0.67

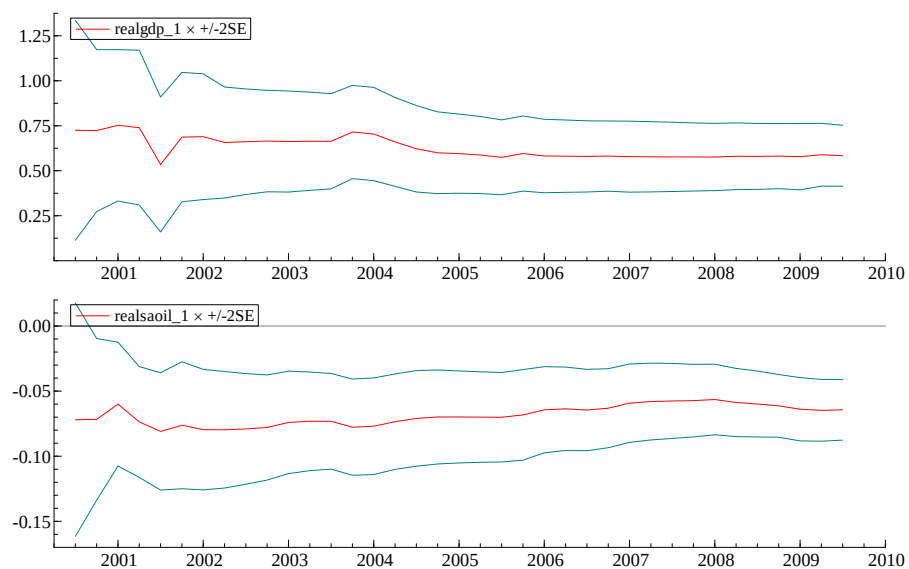
The specific model appears to pass all of the diagnostic tests, except that for heteroscedasticity, as shown in Table 13:

Table 13: Misspecification tests for jet fuel demand specific models

Test name	Test statistic (probability)
AR (1-4) test	0.41 (0.79)
ARCH (1-4) test	2.03 (0.11)
Normality test	0.17 (0.92)
Heteroscedasticity test	2.27 (0.04)
Ramsey RESET	0.31 (0.73)

However, the recursive results suggest that the model produces extremely stable parameter estimates for long-run price and income elasticity, as shown in Figure 6.

Figure 6: Recursive estimates for long-run income elasticity parameter in jet fuel demand models (1998Q1-2009Q3)



Given the stability of the specific model, we perform the bounds test and finds evidence of a statistically significant long-run relationship at a 5% critical level, as shown in Table 14:

Table 14: Bounds test results for jet fuel demand models

Sample period	F-statistic	5% critical bounds	
		Lower bound ($I(0)$)	Upper bound ($I(1)$)
1998Q1-2009Q3	16.11*	3.79	4.85

* Reject null hypothesis at 5% significance level

Similar to the diesel fuel demand, the specific model suggests that jet fuel demand has a fairly low long-run price elasticity of around -0.1 and an income elasticity of about 0.9, as shown in Table 15:

Table 15: Long-run elasticities of jet fuel demand

Sample period	Price elasticity (standard error)	Income elasticity (standard error)
1998Q1-2009Q3	-0.10 (0.01)	0.93 (0.01)

In general, the jet fuel demand function suggests that economic growth, rather than oil prices, is determinative for jet fuel sales in South Africa. In addition, the specific ARDL model indicates a very high speed-of-adjustment parameter (-0.62, see Table 12), which suggests that any long-run disequilibrium is corrected within less than two quarters. The long run relationship, therefore, plays a significant role in quarter-to-quarter consumption changes in South African jet fuel demand.

6. Conclusions

The results from the demand models for petrol, diesel fuel and jet fuel presented in this paper can be summarized as follows. Firstly, petrol demand is more price sensitive than any of the other fuels, with a price elasticity of -0.5. The use of a shorter sample period confirms this elasticity. Secondly, petrol demand is much more income sensitive than previous studies suggest, with an income elasticity of 0.8. Thirdly, both diesel fuel and jet fuel demand has quite low price elasticities of around -0.1 and quite high income elasticities of 1.5 and 0.9 respectively. One should take care when interpreting these three conclusions, especially for petrol demand. Although, in an absolute sense, price elasticity is smaller than income elasticity it does not imply that income forces necessarily dominate fuel demand. Income growth rarely exceeds 5% quarter-on-quarter, while fuel prices can easily reach variation of 10% or more quarter-on-quarter. Suppose now that long-run price changes by 10% and long-run income declines by 5%, one would find their individual long-run effects virtually the same. Elasticity interpretations should therefore be cognisant of the respective size orders of growth in income and price. Fourthly, we do not find any significant changes in price elasticity of petrol or diesel demand in recent years, but do find evidence of stronger income elasticity in petrol demand. Nevertheless, this is not necessarily a vindication of previous annual models, as the structural breaks in our longer-period models suggest that these models' estimates may be due to chance. A final conclusion relates to the speed-of-adjustment parameters in fuel demand models. For the 1998Q1-2009Q3 period the parameter is about -0.25 for petrol demand and around -0.6 for diesel fuel and jet fuel. This suggests that petrol demand takes much longer (about four quarters or more) to adjust to a disequilibrium shock, whereas diesel fuel and jet fuel respond within less than two quarters. This is much shorter than the suggested speed-of-adjustment in previous studies: Akinboade et al (2008) finds -0.42 in their annual model, suggesting an adjustment period of about two and a half years, i.e. 10 quarters.

In general, we argue that our demand models, based on quarterly and shorter sample periods, better explain the dynamic responses of fuel consumption to price and income changes. Although we find similar estimates for specifically petrol demand to those of previous studies when we use a longer

sample period, our research suggests that longer-period models are less stable due to structural breaks and should be avoided.

References

- Akinboade, O. A., E. Ziramba, et al. (2008). "The demand for gasoline in South Africa: an empirical analysis using co-integration techniques." Energy Economics **30**: 3222-3229.
- Bardsen, G. (1989). "Estimation of long-run coefficients in error-correction models." Oxford Bulletin of Economics and Statistics **51**(3): 345-350.
- Breunig, R. V. and C. Gisz (2009). "An exploration of Australian petrol demand: unobservable habits, irreversibility and some updated estimates." The Economic Record **85**(268): 73-91.
- Bureau for Economic Research (2003). The feasibility of a fuel tax levy in the Western Cape. Cape Town, Western Cape Provincial Treasury.
- Cloete, S. A. and E. v. d. M. Smit (1988). "Policy implications of the price elasticity of demand for petrol in South Africa." South African Journal of Science **84**: 227-229.
- De Vita, G., K. Endresen, et al. (2006). "An empirical analysis of energy demand in Namibia." Energy Policy **34**(3447-3463).
- Doornik, J. A. (2008). "Encompassing and automatic model selection." Oxford Bulletin of Economics and Statistics **70**: 915-925.
- Hendry, D. F. and H.-M. Krolzig (2001). Automatic econometric model selection. London, Timberlake Consultants Ltd.
- Li, Z., J. M. Rose, et al. (2010). "Forecasting automobile petrol demand in Australia: an evaluation of empirical models." Transportation Research Part A **44**: 16-38.
- National Energy Regulator of South Africa (2010). Transnet Limited's tariff application for: a 51.3% allowable revenue increase for its petroleum pipelines system for 2010/11. IDMS No. 45374. Pretoria.
- Pesaran, M. H., Y. Shin, et al. (2001). "Bounds Testing Approaches to the Analysis of Level Relationships." Journal of Applied Econometrics **16**: 289-326.
- Theron, N. (2008). "The Sasol/Engen (Uhambo) merger - foreclosure and white fuel demand growth rates." South African Journal of Economic and Management Sciences **11**(3): 264-278.